

Pathogen Genomics Training: Bacterial Genomics & Outbreak Investigation



Bacterial Genomics & Outbreak Investigation Installation Guide

INTRODUCTION

Many bioinformatics tools are developed primarily for Linux. In this course, we will run the tools in a Unix-like environment and install software using **Miniforge**, a lightweight Conda distribution that defaults to the **conda-forge** channel (community-maintained packages with fewer licensing restrictions). This helps ensure participants can reproduce the same software environment across different machines.

Windows users will use **Ubuntu on Windows Subsystem for Linux (WSL)**. **Mac users** can install most tools directly on macOS via Conda/Miniforge, but a few tools may behave differently or may not be available for macOS. If that happens, the course instructor will provide an alternative method.

To learn more about Miniforge and Conda, see: <https://www.fabriziomusacchio.com/blog/2025-07-03-miniforge/> and <https://docs.conda.io/projects/conda/en/latest/user-guide/index.html>

MINIMUM SYSTEM REQUIREMENTS

- **RAM:** 8 GB minimum; 16 GB or more recommended
- **Free storage:** at least 100–200 GB minimum; 500 GB recommended (internal or external drive)
- **Operating system:** Linux, macOS, or Windows with WSL2



8 GB RAM should be sufficient for the workshop dataset. However, 16 GB or more is recommended for smoother performance.

The host-filtering step in module 1 uses a reduced human reference index to keep memory requirements manageable. Workflows using the full human reference genome, larger datasets, or many samples running in parallel may require more RAM and storage.

- ▶ UBUNTU INSTALLATION FOR WINDOWS USERS
- ▶ MINIFORGE INSTALLATION ON UBUNTU FOR WINDOWS USERS
- ▶ MINIFORGE INSTALLATION FOR MAC USERS
- ▶ DOWNLOAD ENVIRONMENT FILES
- ▶ CREATING ENVIRONMENTS AND INSTALLING TOOLS
- ▶ DOWNLOAD FIGTREE

▼ DOWNLOAD THE DATASET

- The dataset (raw fastq files and references) required for the tutorials can be directly downloaded into your course directory. However, the compressed archive is 9.7GB. If you run into issues, your on-site instructors will provide an alternative.
- Download the dataset using wget with -c option to resume download of the partial file if connection breaks.

```
while ! wget -c "https://mendel3.bii.a-star.edu.sg/METHODS/HEIDI/nepal_pgt2026/typhi-workshop.tar.gz";  
do echo "Connection crashed. Retrying in 5 seconds..."; sleep 5; done
```

- Extract the archive

```
tar -xvzf typhi-workshop.tar.gz
```

- Navigate to the decompressed directory

```
cd typhi-workshop
```

- For module 3, you will also require a `metadata.csv` file as well as `gyrA.fasta` reference. Download this with

```
wget https://mendel3.bii.a-star.edu.sg/METHODS/HEIDI/nepal_pgt2026/gyrA.fasta & wget  
https://mendel3.bii.a-star.edu.sg/METHODS/HEIDI/nepal_pgt2026/metadata.csv
```

- Check if you have successfully downloaded all the necessary files

```
# Check if you have two directories: Typhi_fastq and ref as well as two files: gyrA.fasta and  
metadata.csv ls -l # Check that you have 78 fastq files downloaded in Typhi_fastq/ ls -l  
Typhi_fastq/*.gz | wc -l # Check that you have 2 files: CT18_reference.fa and GRCh38_subset.mmi in  
ref/ ls -l ref/
```

- Congratulations! You have successfully completed the installation.